

Subject: Manual, Two step FVE method

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Date : 2026-10-06 20:48:06

1 Description of 1D input file

Five observation points are hard coded in the software, these are located at:

- 1 the centre of the grid.
- 2 at the west boundary.
- 3 at the east boundary.
- 4 halfway to the west boundary.
- 5 halfway to the east boundary.

Extra observation points can be specified in the input file according the description in [item 2](#)

1.1 File description Burgers & Advection-diffusion

An example input file is given in [section 3](#).

Table 1: Standard input-file for Burgers and advection-diffusion equation with default settings.

Keyword	Default	Description
logging	"None"	"iterations" "matrix" "pattern"
[Boundary]		
treg	150.0	Regularization time boundary signal
eps_bc_corr	0.01	Boundary correction term to force the boundary value to the given boundary value
bc_signals	[· , ·]	["constant" "sine"]
bc_type	[· , ·]	["dirichlet" "borsboom"]
bc_vals	[· , ·]	[Real value]
[Domain]		
Lx	-	Length of the domain
x_begin	-	Most left x -coordinate
[Initial]		
ini_vals	[· , ·]	[real value, real value] initial values at west and east boundary
ini_vars	"linear"	["linear", "smoothstep"]

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Keyword	Default	Description
[Numerics]		
dt	5.0	Time step size [s], if dt == 0: then stationary simulation
theta	0.5	Implicitness factor ($0.5 \leq \theta \leq 1.0$)
c_psi	4.0	Smoothness factor used for regularization
iter_max	25	Maximum number of nonlinear iterations
eps_newton	1e-12	Accuracy of the Newton solver (non-linear)
eps_bicgstab	1e-12	Accuracy of the linear solver
eps_abs_function	0.01	Smoothing factor for the absolute-function
linear_solver	"bicgstab"	Type of linear solver "bicgstab", "multigrid"
regularization_init	false	true, false
regularization_iter	false	true, false
regularization_time	false	true, false
[Output]		
dt_his	dt	Time interval to write the history file, [s]
dt_map	60.0	Time interval to write the map file, [s]
dt_screen	60.0	Time interval to write the time step to the screen, [s]
[Physics]		
u_advection	0.0	Advection velocity, [m s^{-1}]
do_convection	false	true, false
do_viscosity	false	true, false
viscosity	0.01	Diffusion/Viscosity coefficient
[Time]		
tunit	"s"	Unit of the start and stop time. Values seconds "s", minutes "m", hours "h" and days "d"
start	0.0	Start time of the simulation, [tunit]
stop	60.0	Stop time of the simulation, [tunit]
[[ObservationPoint]]		
x	-	x -coordinate of the observation point
y	-	y -coordinate of the observation point
name	"<name>"	Name of observation point between double quotes

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Keyword	Default	Description
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1.2 File description

Wave 1D

An example input file is given in [section 4](#).

Table 2: Standard input-file for wave-1d equation with default settings.

Keyword	Default	Description
logging	"None"	"iterations" "matrix" "pattern"
[Boundary]		
treg	150.0	Regularization time boundary signal
eps_bc_corr	0.01	Boundary correction term to force the boundary value to the given boundary value
bc_type	[· , ·]	["borsboom" "mooiman"]
bc_vals	[· , ·]	[Real values]
bc_vars	[· , ·]	["zeta" "q"]
[Domain]		
Lx	-	Length of the domain, [m]
depth	10.0	Overruled by the keyword <code>bed_level_file</code> Bathymetry below reference level [m]
geometry_type	"<name>"	Overruled by the keyword <code>bed_level_file</code> "UniformGeometry" Uniform depth profile "WavyBedGeometry" Wavy bed geometry "SlopedBedGeometry" Sloped bed geometry (Chézy flow) "SlopedWavyBedGeometry" Sloped wavy bed geometry "WeirBorsboom2000Geometry" Profile of a weir "Shoal" Profile of a shoal
bed_level_file	"<name>"	Overrules keywords <code>depth</code> and <code>geometry_type</code> Name of bed-level-file between double quotes
[Initial]		
ini_vars	"-"	["zeta_gauss_hump"]
gauss_app	"-"	Twice the amplitude of the Gaussian hump at the boundary
gauss_mu	"-"	Location of the top of the Gaussian hump
gauss_sigma	"-"	Width of the Gaussian hump

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Keyword	Default	Description
[Numerics]		
dt	-	Time step size [s], if dt==0 then stationary simulation
dx	-	Grid size, [m]
theta	0.5	Implicitness factor ($0.5 \leq \theta \leq 1.0$)
c_psi	4.0	Smoothness factor used for regularization
iter_max	25	Maximum number of nonlinear iterations
eps_newton	1e-12	Accuracy of the Newton solver (non-linear)
eps_bicgstab	1e-12	Accuracy of the linear solver
eps_abs_function	0.01	Smoothing factor for the absolute-function
linear_solver	"bicgstab"	Type of linear solver "bicgstab", "multigrid"
regularization_init	false	true, false
regularization_iter	false	true, false
regularization_time	false	true, false
[Output]		
dt_his	dt	Time interval to write the history file, [s]
dt_map	60.0	Time interval to write the map file, [s]
dt_screen	60.0	Time interval to write the time step to the screen, [s]
[Physics]		
do_convection	false	true, false
do_viscosity	false	true, false
viscosity	0.01	Diffusion/Viscosity coefficient
do_source	false	true, false
src_type	"none"	"none", "quadratic"
[Time]		
tunit	"s"	Unit of the start and stop time. Values seconds "s", minutes "m", hours "h" and days "d"
start	0.0	Start time of the simulation, [tunit]
stop	60.0	Stop time of the simulation, [tunit]
[[ObservationPoint]]		
x	-	x -coordinate of the observation point
y	-	y -coordinate of the observation point
name	"<name>"	Name of observation point between double quotes

2 Description of 2D input file

Nine observation points are hard coded in the software, these are located at:

- 1 the centre of the grid.
- 2 in the middle of the north boundary.
- 3 in the middle of the east boundary.
- 4 in the middle of the south boundary.
- 5 in the middle of the west boundary.
- 6 in the North-East corner of the grid.
- 7 in the South-East corner of the grid.
- 8 in the South-West corner of the grid.
- 9 in the North-West corner of the grid.

Extra observation points can be specified in the input file according the description in section [File description](#).

File description

Items in the color red are only to be used in the development stage of the software. An example input file is given in [section 5](#).

Table 3: Standard input-file with default settings.

Keyword	Default	Description
logging	"None"	"iterations" "matrix" "pattern"
[Boundary]		
treg	150.0	Regularization time boundary signal
eps_bc_corr	0.01	Boundary correction term to force the boundary value to the given boundary value
bc_type	[· , · , · , ·]	["free-slip" "no-slip" "borsboom" "mooiman"]
bc_vars	[· , · , · , ·]	["-" "zeta" "q"]
bc_vals	[· , · , · , ·]	[Real value]
[Domain]		
bed_level_file	"<name>"	Name of bed-level-file between double quotes
mesh_file	"<name>"	Name of mesh file point between double quotes

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Keyword	Default	Description
[Initial]		
ini_vars	None	["zeta", "zeta_gauss_hump" "zeta_constant" "zeta_gauss_hump_x" "zeta_gauss_hump_y"]
gauss_amp	0.0	
gauss_mu	0.0	
gauss_mu_x	0.0	
gauss_mu_y	0.0	
gauss_sigma	350.0	
gauss_sigma_x	350.0	Optional if "gauss_sigma" is not specified
gauss_sigma_y	350.0	Optional if "gauss_sigma" is not specified
[Numerics]		
dt	5.0	Time step size [s], if dt==0 then stationary simulation
theta	0.5	Implicitness factor (0.5 <= theta <= 1.0)
c_psi	4.0	Smoothness factor used for regularization
iter_max	25	Maximum number of nonlinear iterations
eps_newton	1e-12	Accuracy of the Newton solver (non-linear)
eps_bicgstab	1e-06	Accuracy of the linear solver
eps_abs_function	0.01	Smoothing factor for the absolute-function
linear_solver	"bicgstab"	Type of linear solver "bicgstab", "multigrid"
regularization_init	false	true, false
regularization_iter	false	true, false
regularization_time	false	true, false
[Physics]		
do_linear_waves	true	true, false
do_continuity	true	true, false
do_q_equation	true	true, false
do_r_equation	true	true, false
do_convection	false	true, false
do_bed_shear_stress	false	true, false
do_viscosity	false	true, false
chezy_coefficient	25.0	Bed friction coefficient
viscosity	0.01	Diffusivity coefficient
[Output]		
dt_his	dt	Time interval to write the history file, [s]
dt_map	60.0	Time interval to write the history file, [s]
dt_screen	60.0	Time interval to write the time step to the screen, [s]
[Time]		
tunit	"s"	Unit of the start and stop time. Values seconds "s", minutes "m", hours "h" and days "d"
start	0.0	Start time of the simulation, [s]
stop	60.0	Stop time of the simulation, [s]

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Keyword	Default	Description
[[ObservationPoint]]		
x	-	x -coordinate of the observation point
y	-	y -coordinate of the observation point
name	"<name>"	Name of observation point between double quotes

3 Example input 1D file

Advection-diffusion

```

Logging = "None" # "iterations", "matrix", "pattern"

[Boundary] # west, east
treg      = 1800.0 # Regularization time boundary signal [s]
eps_bc_corr = 0.01
bc_signals = ["constant", "constant"]
bc_type    = ["dirichlet", "borsboom"]
bc_vals    = [2.0, 0.0]

[Domain]
Lx      = 20000.0 # Domain length [m]
x_origin = 0.0 # Origin of the x-coordinate [m]

[Initial]
ini_vals = [1.0]
ini_vars = ["constant"]

[Numerics]
dt      = 1.0 # Time step size [s], if dt == 0: then stationary problem
dx      = 1.0 # Grid size [m]
c_psi   = 4.0
theta   = 0.50 # Implicitness factor (0.5 <= theta <= 1.0)
iter_max = 25.0 # Maximum number of non-linear iterations
eps_newton = 1e-12
eps_bicgstab = 1e-12
eps_abs_function = 0.01
regularization_init = true
regularization_iter = false
regularization_time = true

[Output]
dt_his    = 1.0
dt_map    = 300.0
dt_screen = 300.0

[Physics]
do_convection = true
do_viscosity  = true
viscosity     = 0.1
do_source     = false
src_type      = "none"

[Time]
tunit = "s" # "s", "m", "h", "d"
tstart = 0.0
tstop  = 14400.0

# Hard coded observation points are not listed

[[ObservationPoint]]
x      = 2500
y      = 0
name   = "_a"

[[ObservationPoint]]

```

```
x      = 7500
y      = 0
name = "_b"

[[ObservationPoint]]
x      = 12500
y      = 0
name = "_c"

[[ObservationPoint]]
x      = 17500
y      = 0
name = "_d"
```

4 Example input 1D file Wave 1D

```
[Boundary]
bc\_type = ["borsboom", "borsboom"] # Type "mooiman", "borsboom"
bc\_vars = ["q", "zeta"]             # Incoming signal "q" or "zeta"
bc\_vals = [0.0, 0.0]                # The inflow discharge
treg = 600.0                         # Regularization time on boundary
eps\_bc\_corr = 1.0e-2

[Domain]
Lx = 36000.0 # Domain length [m]
depth = 10.0 # bathymetry below reference level [m]
geometry\_type = "UniformGeometry"

[Initial]
ini\_vars = ["zeta\_gauss\_hump"]
gauss\_mu = 9000.0 # location of the top of the gaussian hump
gauss\_sigma = 700.0 # location of the top of the gaussian hump
gauss\_amp = 0.02 # twice the amplitude of the gaussian hump at the boundary

[Numerics]
dt = 10.0 # Time step size [s], if dt == 0: then stationary problem
dx = 10.0 # Grid size [m]
iter\_max = 25 # Maximum number of nonlinear iterations
theta = 0.501 # Implicitness factor (0.5 <= theta <= 1.0)
eps\_newton = 1.0e-12
eps\_bicgstab = 1.0e-12

[Output]
dt\_his = 10.0 # [s]
dt\_map = 1200.0 # [s]
dt\_screen = 300.0 # [s]

[Physics]
g = 10.0 # Gravitational acceleration
do\_convection = false # Include momentum convection term

[Time]
tstart = 0.0 # Start time [s]
tstop = 43200.0 # End time [s]
```

5 Example input 2D file

```

logging = "None" # "iterations", "matrix", "pattern"

[Boundary] # north, east, south, west
treg      = 150.0 # Regularization time boundary signal
eps_bc_corr = 0.01
bc_type    = ["borsboom", "borsboom", "borsboom", "borsboom"] # Type "neumann", "dirichlet", "borsboom"
bc_vars    = ["zeta", "zeta", "zeta", "zeta"]
bc_absorbing = [true, true, true, true]
bc_vals    = [0.0, 0.0, 0.0, 0.0]

[Domain]
mesh_file    = "geometry/62x62_6x6km_net.nc"
bed_level_file = "geometry/62x62_6x6km_flat.dep"

[Initial]
ini_vars     = ["zeta", "zeta_gauss_hump"]
Gauss_amp    = 1.0
Gauss_mu     = 0.0
Gauss_mu_x   = 0.0
Gauss_mu_y   = 0.0
Gauss_sigma  = 350.0
Gauss_sigma_x = 350.0
Gauss_sigma_y = 350.0

[Numerics]
dt           = 5.0 # Time step size [s], if dt == 0: then stationary problem
theta        = 0.501 # Implicitness factor (0.5 <= theta <= 1.0)
c_psi        = 4.0
iter_max     = 25 # Maximum number of nonlinear iterations
eps_newton   = 1e-12
eps_bicgstab = 1e-06
eps_abs_function = 0.01
linear_solver = "bicgstab" # "bicgstab", "multigrid"
regularization_init = false
regularization_iter = false
regularization_time = false

[Physics]
do_linear_waves = true
do_continuity   = true
do_q_equation   = true
do_r_equation   = true
do_convection   = false
do_bed_shear_stress = false
do_viscosity    = false
chezy_coefficient = 25.0
viscosity       = 100.0

[Output]
dt_his = 0.01
dt_map = 30.0

[Time]
tunit = "s"
tstart = 0.0
tstop = 1800.0

```

